

BRITISH AIRWAYS

Passenger Satisfaction & Operations Executive Insights

Data-Driven Fleet and Service Optimization

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Project Overview & Architecture

CONTEXT, BUSINESS OBJECTIVE & ANALYTICS FRAMEWORK

| The Business Dilemma

- **Fragmented Feedback:** British Airways captures huge volumes of customer reviews spanning multiple categories (Value, Seat Comfort, Ground Service).
- **The Outlier Challenge:** Raw mathematical averages were heavily skewed by low-frequency aircraft types with isolated single 10/10 or 1/10 reviews, distorting strategic asset decisions.

| Project Scope & Objective

- Develop a centralized, interactive analytics canvas to isolate core operational performance.
- Empower executive stakeholders to seamlessly toggle between operational metrics with complete integrity of historical trends.

| Technical Toolstack & Methods

- **Power BI Desktop:** Core enterprise environment for end-to-end extraction, transformations, data modeling, and UI layout orchestration.
- **DAX (Data Analysis Expressions):** Used advanced dynamic context shifting using SWITCH and SELECTEDVALUE measures to enable single-card multi-metric swapping.
- **Advanced Visual Filtering:** Implemented visual-level frequency parameters (`Total Reviews ≥ 5`) to dynamically filter out statistical anomalies.
- **Geospatial Intelligence:** Mapped international passenger sentiment metrics globally via Bing Maps integration.

Executive KPI Performance Overview

HIGH-LEVEL BRAND SENTIMENT VS. CORE OPERATIONAL FRICTION

2.80 Value for Money	2.90 Seat Comfort	3.11 Ground Service	4.39 Overall Rating
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Key Insights Summary

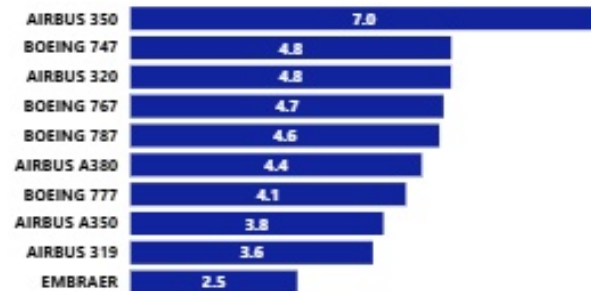
- **The Core Disconnect:** While the overall brand perception sits healthy at **4.39 / 5.00**, the individual core baseline scores flag severe underlying issues.
- **Physical Product Pricing Friction:** **Value for Money (2.80)** and **Seat Comfort (2.90)** are critically low, identifying that passengers do not feel the onboard hardware matches the premium fare structure.
- **Ground Operations Resilience:** Ground Service stands tall at **3.11**, proving terminal, lounge, and gate operations outpace the cabin experience.



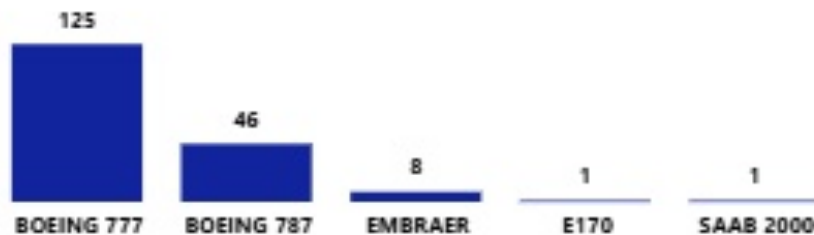
Fleet Performance Evaluation

RECALIBRATING RATINGS VIA STATISTICAL THRESHOLD OPTIMIZATION

Fleet Performance Evaluation



Total Review Count by aircraft



Data Refinement & Findings

- **Eliminating Outlier Noise:** Initial raw data placed the regional E170 fleet at a perfect 10.0 average due to a single isolated customer review.
- **Applying the Threshold Filter:** Enforcing a minimum volume constraint (Total Reviews ≥ 5) safely eliminated low-frequency noise.
- **True Fleet Hierarchy Revealed:** The long-haul operational workhorses now sit correctly ranked:
 - **Airbus A350:** Leading premium tier (7.0 rating index).
 - **Boeing 747:** Highly stable legacy performer (4.8 score).
 - **Boeing 777:** Holds the ultimate baseline with **125 total reviews** at a 4.1 rating.

Temporal Trends & Geospatial Distribution

SEASONALITY IMPACTS AND GLOBAL SENTIMENT FOOTPRINTS

Passenger Satisfaction Trend Analysis



Clear cyclical spikes appear around **May** and **September** (surpassing 5.0), matching seasonal premium travel windows, followed by distinct dips in March and October.

Global Sentimental Heat Map



Dense feedback concentrations across **Europe (London Hub)** and key transatlantic corridors in **North America** outline the critical financial pathways requiring strict quality controls.

Strategic Action Plan & Conclusion

TRANSLATING POWER BI VISUAL INTELLIGENCE INTO CORPORATE VALUE

| 1. High-Volume Fleet Refurbishment (The Boeing 777 Focus)

- **The Logic:** Because the ****Boeing 777**** commands the maximum feedback weight (125 reviews), minor cabin upgrades will create the highest overall boost to fleet satisfaction.
- **Action:** Prioritize physical seat-comfort overhauls and legroom modifications on this fleet to address the sub-3.0 seat rating immediately.

| 2. Value Realignment Strategy

- **The Logic:** The deep gap between ****Overall Brand Impression (4.39)**** and ****Value for Money (2.80)**** shows an economic pricing mismatch.
- **Action:** Review long-haul pricing structures and introduce premium targeted seat selection credits to offset the value-friction point.

| 3. Leverage Operational Strengths

- **The Logic:** Ground service efficiency (3.11) is an existing corporate competitive edge.
- **Action:** Seamlessly market premium, frictionless terminal lounge pathways as part of long-haul campaigns to mitigate onboard seating complaints.